

Member ID: _____

Time: _____

Rank: _____



Network Administration Using Cisco (315)

State 2024

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each) _____ (100 points)

Test Time: 60 minutes

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. What does the TCP protocol in the OSI model primarily manage?
 - A. The physical transmission of the data
 - B. The routing of data across different networks
 - C. Ensuring reliable data transmission by managing data packet error checking and sequencing
 - D. The direct communication between software applications
2. What is the purpose of a VLAN in a network?
 - A. It is used to redistribute routes from other routing protocols.
 - B. It is a subset of a network that groups together devices from different physical LANs.
 - C. It connects devices within a LAN through IP addressing.
 - D. It is a firewall used to protect the network from external threats.
3. What is the main function of EIGRP in network routing?
 - A. It is used for local network management.
 - B. It establishes secure VPN tunnels.
 - C. It distributes IP addresses within a network.
 - D. It is an advanced distance-vector routing protocol, used on a computer network for automating routing decisions and configuration.
4. What is the primary role of a Router in a network?
 - A. It translates IP addresses to domain names
 - B. It provides the graphical user interface for network management
 - C. It forwards data packets between computer networks
 - D. It controls the access of computers to the network
5. What is the primary role of a Dynamic Host Configuration Protocol (DHCP) server in a network?
 - A. It distributes summarized routes of the area to other OSPF areas
 - B. It translates domain names to IP addresses
 - C. It assigns IP addresses to devices on a network
 - D. It controls the access of computers to the network
6. What is the function of the Simple Network Management Protocol (SNMP)?
 - A. It provides a set of standards for network management, including monitoring and controlling network devices
 - B. It is always the Designated Router of the OSPF area
 - C. It always runs other routing protocols to translate them into the OSPF area
 - D. It distributes summarized routes of the area to other OSPF areas

7. What is the purpose of an Intrusion Detection System (IDS) in a network?
 - A. It acts as a default gateway for the network
 - B. It monitors network and/or system activities for malicious activities or policy violations
 - C. It assigns IP addresses to devices on a network
 - D. It translates IP addresses to domain names
8. What does a Virtual Local Area Network (VLAN) do in a network?
 - A. It encrypts all data traffic for enhanced security
 - B. It physically separates devices into different geographic areas
 - C. It logically separates devices on the same network into different subnetworks
 - D. It distributes summarized routes of the area to other OSPF areas
9. What is a purpose of Network Address Translation (NAT) in a network?
 - A. It assigns IP addresses dynamically to devices on a network
 - B. It manages the routing tables in an OSPF area
 - C. It forwards data packets based on MAC addresses
 - D. It converts private IP addresses into public IP addresses for outbound traffic
10. Which interfaces would be impacted by the command "network 172.16.0.0 0.0.255.255 area 0" in OSPF?
 - A. All interfaces on the router that have an IP address in the range 172.16.0.0 to 172.16.255.255
 - B. All interfaces on the router that have an IP address outside the range 172.16.0.0 to 172.16.255.255
 - C. All interfaces on the router that have an IP address in the form of x.x.x.0 (X indicates wildcards)
 - D. All interfaces on the router that have an IP address in the form of x.x.x.255 (X indicates wildcards)
11. Which command sets a Cisco switch interface to trunk mode?
 - A. switchport mode trunk
 - B. switchport trunk mode
 - C. switchport set trunk
 - D. trunk mode switchport
12. How can you set the native VLAN for a trunk link to VLAN 50?
 - A. switchport trunk native vlan 50
 - B. switchport mode native vlan 50
 - C. switchport native vlan 50
 - D. native vlan 50
13. Which command should you use to remove VLAN 10 from a Cisco switch?
 - A. delete vlan 10
 - B. remove vlan 10
 - C. no vlan 10
 - D. unset vlan 10

14. What does the command "show ip route" do on a Cisco router?
 - A. It shows the entire IP routing table.
 - B. It shows only the active IP routes.
 - C. It shows the MAC address table.
 - D. It shows the routing protocols configured on the router.
15. What is the result of running the command "show vlan brief" on a Cisco switch?
 - A. It shows a brief summary of all VLANs and their associated interfaces.
 - B. It shows the detailed configuration of all VLANs.
 - C. It shows only the active VLANs.
 - D. It shows the detailed information for a specific VLAN.
16. What commands set and enable an IP Address on a FastEthernet interface?
 - A. ip address 192.168.10.2 255.255.255.0, enable
 - B. interface FastEthernet 0/1, ip address 192.168.10.2 255.255.255.0, no shutdown
 - C. interface FastEthernet 0/1, address 192.168.10.2, enable
 - D. FastEthernet address 192.168.10.2 255.255.255.0, no shutdown
17. What command set is used to set up a static route to a particular network?
 - A. ip route 192.168.10.0 255.255.255.0 192.168.1.1, enable
 - B. ip route 192.168.10.0 255.255.255.0 192.168.1.1, no shutdown
 - C. ip route 192.168.10.0 192.168.1.1, enable
 - D. ip route 192.168.10.0 255.255.255.0 192.168.1.1
18. How to set the clock rate on a Serial interface?
 - A. interface Serial 0/0, clock rate 64000, enable
 - B. interface Serial 0/0, clock rate 64000
 - C. clock rate 64000, enable
 - D. Serial 0/0, clock rate 64000
19. How to set and enable Port Security on a FastEthernet interface?
 - A. interface FastEthernet 0/1, switchport port-security, enable
 - B. switchport port-security, enable
 - C. interface FastEthernet 0/1, port-security, enable
 - D. interface FastEthernet 0/1, switchport port-security, no shutdown
20. Which of the following IP addresses is in the same IPv4 subnet as 192.168.16.24/27?
 - A. 192.168.16.5
 - B. 192.168.16.30
 - C. 192.168.17.26
 - D. 192.168.16.90
21. Which of the following addresses belongs to the same IPv4 subnet as 172.16.10.64/26?
 - A. 172.16.10.100
 - B. 172.16.11.67
 - C. 172.16.10.63
 - D. 172.16.9.65

22. Which of the following is a private Class A network address?
- A. 10.5.0.0/16
 - B. 165.15.0.0/16
 - C. 172.31.0.0/16
 - D. 191.168.0.0/16
23. What is the broadcast address for the IP 10.45.100.28/27?
- A. 10.45.100.31
 - B. 10.45.100.59
 - C. 10.45.100.63
 - D. 10.45.100.127
24. Your network uses the address range 10.0.0.0/8 and needs to be subdivided into at least 600 individual networks that will allow as many hosts as possible on each. Which subnet mask should you use?
- A. 255.255.128.0
 - B. 255.255.248.0
 - C. 255.255.240.0
 - D. 255.255.192.0
25. How would you verify the IPv6 configuration on a Cisco device?
- A. show ipv6 configuration
 - B. show ipv6 interface
 - C. ipv6 show routing
 - D. show ipv6 route
26. How can you disable IPv6 on a specific interface in a Cisco device?
- A. ipv6 disable
 - B. no ipv6 enable
 - C. no ipv6 address
 - D. disable ipv6
27. What is the primary function of the Cisco technology, Spanning Tree Protocol (STP)?
- A. STP is designed to prevent loop formation in a network.
 - B. STP is used to create multiple virtual LANs on a single physical network.
 - C. STP is a security protocol to prevent unauthorized access to a network.
 - D. STP is a protocol to manage multiple gateways in a network.
28. Identify the IPv6 Link-Local address:
- A. 2001:DB8::1
 - B. 192.168.1.1
 - C. FE80::1
 - D. 172.16.0.1

29. Which command is used to configure a default gateway on a Cisco switch?
- A. ip default-gateway
 - B. ip route 0.0.0.0 0.0.0.0
 - C. ip gateway default
 - D. ip default-route
30. A network administrator wants to minimize the time it takes for a port to transition from blocking to forwarding state in spanning-tree. Which of the following configurations should be implemented?
- A. Enable Rapid Spanning Tree Protocol (RSTP)
 - B. Disable spanning-tree on the port
 - C. Increase the port priority on the switch
 - D. Configure the port as an access port
31. A network administrator wants to decrease the convergence time of spanning-tree in the network. Which feature should be enabled?
- A. Bridge Protocol Data Units (BPDU) guard
 - B. Root guard
 - C. PortFast
 - D. Loop guard
32. The command "show interface serial0/0" was executed on a router, and the following entry is found: "Serial0/0 is up, line protocol is up". What can be determined from this output?
- A. The physical interface Serial0/0 is operational and connected.
 - B. The line protocol on Serial0/0 is in the up state.
 - C. The router is using the HDLC encapsulation protocol on Serial0/0.
 - D. The router is experiencing errors on the physical layer of Serial0/0.
33. When configuring VLAN Trunking Protocol (VTP) in a Cisco network, what is the purpose of the "transparent" mode?
- A. It forwards VTP advertisements but does not synchronize VLAN information
 - B. It forwards VTP advertisements and synchronizes VLAN information
 - C. It blocks VTP advertisements from being forwarded
 - D. It allows VTP advertisements from unknown sources but notates the event by sending a message
34. When configuring access control lists (ACLs), what is the purpose of the "deny" statement?
- A. It allows traffic matching the specified criteria
 - B. It redirects traffic matching the specified criteria to another destination
 - C. It blocks traffic matching the specified criteria
 - D. It prioritizes traffic matching the specified criteria over other traffic

35. In the context of Dynamic Host Configuration Protocol (DHCP), what is the purpose of the "exclusion" range?
- A. It blocks all incoming DHCP requests
 - B. It allows incoming DHCP requests from unknown sources but notates the event by sending a message
 - C. It prevents specific IP addresses from being assigned by the DHCP server
 - D. It allows specific IP addresses to be assigned by the DHCP server
36. Which command is used to configure a default gateway on a Cisco switch?
- A. switch default-gateway
 - B. default-gateway ip
 - C. gateway default
 - D. ip default-gateway
37. Which command is used to display the current configuration of a Cisco router?
- A. show running-config
 - B. show startup-config
 - C. show configuration
 - D. show config
38. Which command is used to verify the connectivity between two Cisco devices?
- A. traceroute
 - B. ping
 - C. show interfaces
 - D. show cdp neighbors
39. Which command is used to apply an access control list (ACL) to an interface?
- A. ip access-group
 - B. ip access-list
 - C. apply-acl
 - D. apply-access-list
40. In a Cisco ACL, which wildcard mask would match all IP addresses within a given network?
- A. 0.0.0.0
 - B. 255.255.255.0
 - C. 0.0.0.255
 - D. 255.255.255.255
41. Which of the following is true about wildcard masks used in access control lists (ACLs)?
- A. Wildcard masks are used to specify the source and destination IP addresses in an ACL.
 - B. Wildcard masks are used to define the subnet mask in an ACL.
 - C. Wildcard masks are used to identify which bits of an IP address should be ignored in an ACL.
 - D. Wildcard masks are used to specify the action to be taken on packets matching an ACL.

42. How can you verify the DHCP lease information for a specific client on a Cisco router?
- A. show dhcp server lease
 - B. show dhcp binding
 - C. show dhcp client lease
 - D. show dhcp pool
43. What is the default lease time for DHCP IP addresses?
- A. 1 hour
 - B. 1 day
 - C. 7 days
 - D. 30 days
44. What is the default port number used by DHCP?
- A. 20
 - B. 53
 - C. 67
 - D. 80
45. What is the recommended minimum number of NTP servers for reliable time synchronization?
- A. 1
 - B. 2
 - C. 3
 - D. 4
46. What is a firewall?
- A. A hardware device used to connect networks
 - B. A software program that protects against unauthorized access
 - C. A physical barrier preventing network communication
 - D. A protocol used for secure file transfers
47. What is the purpose of a VPN?
- A. To create a private network within a public network
 - B. To increase internet speed
 - C. To block unwanted advertisements
 - D. To encrypt email communications
48. What is the difference between HTTP and HTTPS?
- A. HTTPS is a more secure version of HTTP
 - B. HTTP is used for websites, while HTTPS is used for emails
 - C. HTTPS uses a different port number than HTTP
 - D. HTTP is faster than HTTPS
49. What is the purpose of an IP address?
- A. To identify a specific website on the internet
 - B. To encrypt data for secure transmission
 - C. To track the location of a mobile device
 - D. To uniquely identify a device on a network

50. Which protocol and port combination is commonly used for secure web browsing?
- A. UDP port 80
 - B. TCP port 443
 - C. UDP port 443
 - D. TCP port 80